About this book

A practical introduction for everyone interested in the simulation of high-voltage electrical power networks.

Over the past 25 years, I have developed two educational toolboxes for LTspice. The first supports nonlinear load-flow calculations; the second is designed for the transient simulation of electrical power networks.

The toolboxes contain the essential components required to build realistic power-system models:

- buses and sources;
- synchronous machines and transformers;
- overhead lines and power cables;
- loads and shunt reactors; and
- surge-protection devices, Petersen coils, and many other elements.

Some elements provide parameters that can be changed directly by the user. Others are supplied with carefully selected presets, which can also be overridden whenever a study requires different values.

The transformer library ranges from simple models to sophisticated representations with a nonlinear iron core, hysteresis, and remanent flux. Eight vector groups and voltage levels from 400 kV down to 400 V are included. Additional vector groups can be created by editing the macro-models, because all `.lib` files are freely accessible and editable.

Hundreds of examples for both toolboxes are available on my website, ranging from simple introductory circuits to complex networks. Many schematic files contain their own documentation, and the first nine examples are explained in detail in this book.

These toolboxes are intended especially for students of electrical engineering, but they are equally useful to teachers, engineers, and anyone who wants to explore power-system behaviour through simulation.

I hope you enjoy experimenting with these simulations as much as I do. Your feedback is always welcome.

Friedrich Schmidt